

Upgraded SW Models

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1 Shallow Water Equations with temperature gradient

Integrating the primitive equations vertically, we get that for a layer going from z_1 to z_2 , with vertically averaged density being $\bar{\rho}$:

$$\bar{\rho}(z_2 - z_1)(D_t \mathbf{v} + \mathbf{f} \times \mathbf{v}) = -\nabla \left(-g\bar{\rho} \frac{(z_2 - z_1)^2}{2} + (z_2 - z_1)p_{z_1} \right) - \nabla z_1 p_{z_1} + \nabla z_2 p_{z_2} \quad (1)$$

Note that here we used

$$p_z = -g\bar{\rho}(z - z_1) + p_{z_1} \quad (2)$$

We can write it in terms of geopotential $\phi = -g \frac{\bar{\rho}}{\rho_0}(z - z_1) + \phi_{z_1}$ and buoyancy as

$$(z_2 - z_1)(D_t \mathbf{v} + \mathbf{f} \times \mathbf{v}) = -\nabla \left(-g \frac{\bar{\rho}}{\rho_0} \frac{(z_2 - z_1)^2}{2} + (z_2 - z_1)\phi_{z_1} \right) - \nabla z_1 \phi_{z_1} + \nabla z_2 \phi_{z_2} \quad (3)$$

The integration of continuity equation, combined with incompressibility condition $\nabla \cdot \mathbf{v} = 0$ and mean field hypothesis $\langle \rho \mathbf{v} \rangle = \langle \rho \rangle \langle \mathbf{v} \rangle$, similar to normal SW, becomes

$$\partial_t(z_2 - z_1) + \nabla \cdot ((z_2 - z_1)\mathbf{v}) = 0 \quad (4)$$

And the column-averaged density is a materially conserved tracer

$$D_t \bar{\rho} = 0 \quad (5)$$

2 Adding Moisture into the SW Equations

Again we start from primitive equations:

$$D_t \mathbf{v} + \mathbf{f} \times \mathbf{v} = -\nabla \Phi \quad (6)$$

$$\nabla \mathbf{v} + \partial_z w = 0 \quad (7)$$

$$D_t(\theta + \frac{L}{C_p}q) = 0 \quad (8)$$

And if there is no condensation, θ and q are separately conserved. Now considering a two-layer structure, with the bottom layer based at z_0 and top layer at z_1 , top at z_2 .

$$w_0 = \frac{dz_0}{dt}, w_1 = \frac{dz_1}{dt} + W_1, w_2 = \frac{dz_2}{dt} + W_2 \quad (9)$$

So the continuity equations write

$$\partial_t h_1 + \nabla \cdot (h_1 \mathbf{v}_1) = W_0 - W_1 \quad (10)$$

$$\partial_t h_2 + \nabla \cdot (h_2 \mathbf{v}_2) = W_1 - W_2 \quad (11)$$

The vertically averaged momentum equation then becomes

$$\partial_t \mathbf{V}_1 + (\mathbf{v}_1 \cdot \nabla) \mathbf{V}_1 + \mathbf{f} \times \mathbf{V}_1 = -\nabla \Phi|_{z_1} + g \frac{\theta_1}{\theta_0} \nabla z_1 \quad (12)$$

$$\partial_t \mathbf{V}_2 + (\mathbf{V}_2 \cdot \nabla) \mathbf{V}_2 + \mathbf{f} \times \mathbf{V}_2 = -\nabla \Phi|_{z_2} + g \frac{\theta_2}{\theta_0} \nabla z_2 + \frac{v_1 - v_2}{h} W_1 \quad (13)$$

Here if we take $\phi_0 = 0$, $\phi_1 = g \frac{\theta_1}{\theta_0} (z_1 - z_0)$, and $\phi_2 = g \frac{\theta_1}{\theta_0} (z_1 - z_0) + g \frac{\theta_2}{\theta_0} (z_2 - z_1)$. Taking the vertically integrated water content in layer i be Q_i , we have a local conservation

$$\partial_t Q_i + \nabla \cdot (Q_i \mathbf{V}_i) = E_i - P_i \quad (14)$$

Here E and P are the evaporation and precipitation respectively, which feeds in the forcing of the model. The vertical speed at the interfaces are directly related to the precipitation rate

$$W_i = \beta_i P_i, \beta_i = \frac{L}{C_p(\theta_2 - \theta_1)} \quad (15)$$

Under the assumption of a stably stratified background and a dry layer 2, only layer 1 has precipitations, we have

$$\beta_1 \approx \frac{1}{q_1} > 0 \quad (16)$$

Here we can use a relaxation scheme to calculate the precipitation rate, using the saturated $Q_s = q_s(P, T)h$, or $P_1 = \tau_1^{-1}(Q - Q_s)$, we use a realistic value of P and T to obtain the saturated vapor mixing ratio at this point. In addition, we need a water mixing ratio q_0 , which is assumed to be a uniform distributed in the abyssal layer. What is the speed that precipitation is evaporated and fed back into the weather layer again, or, how do we calculate W_0 ? We assume all the precipitation falls into the abyssal layer and averaged, and that the evaporation rate is proportional to this precipitation,

$$\frac{d\alpha_0}{dt} = \bar{P}_1 - \tau_2^{-1}\alpha_0 \quad (17)$$

And the speed means the same amount of the moisture is fed back to layer 1 from layer 0, so that

$$W_0 = \frac{\alpha_0}{\tau_2 q_0} \quad (18)$$

This also gives the value of E , being

$$E = W_0 q_0 \quad (19)$$

Thus we close the system. Still, in addition to an infinitely deep lower layer (abyssal layer), we would like to note that the upper layer has very small impact on the lower layer, especially when we ignore the momentum transfer between these two layers. Ignoring the highest layer, just with the weather layer, we can have a one-layer model for the water. The forcing can be simply applied to its layer thickness after we have calculated the values of W_0 and W_1 .

The one-layer model with moisture This model has a one-layer setup. The physical quantities include:

- The layer height H and the column-averaged horizontal velocity $\mathbf{v}$.
- The column integrated water content in layer 1, Q_1 , and the water content in the abyssal layer, α_0 .
- The local upward velocities at the bottom and top boundaries of this layer, W_0 and W_1 .

Note that the above-listed variables imply several other quantities used (potentially as parameters) but not set as variables:

- The potential temperature values θ_0 , θ_1 , and θ_2 , and gravity g .
- The time constants τ_1 and τ_2 , controlling the precipitation and evaporation, respectively.
- The water contents in the abyssal layer q_0 and the saturated vapor concentration in the weather layer q_s .

And the equations that this shallow water model follows:

$$\partial_t H + \nabla \cdot (H\mathbf{v}) = W_0 - W_1 \quad (20)$$

$$\partial_t (H\mathbf{v}) + (\mathbf{v} \cdot \nabla)(H\mathbf{v}) + \mathbf{f} \times (H\mathbf{v}) = g \frac{\theta_1}{\theta_0} \nabla z_0 \quad (21)$$

$$W_1 = \frac{LH(q - q_s)}{\tau_1 C_p (\theta_2 - \theta_1)} \quad (22)$$

$$W_0 = \frac{\alpha_0}{\tau_2 q_0} \quad (23)$$

$$\partial_t (Hq) + \nabla \cdot (Hq\mathbf{V}_i) = W_0 q_0 - \frac{H(q - q_s)}{\tau_1} \quad (24)$$

$$\frac{d\alpha_0}{dt} = \frac{\int_A H(q - q_s) dA}{\tau_1 \int_A dA} - \tau_2^{-1} \alpha_0 \quad (25)$$

3 The two-layer coupled moist convection structures

Thinking about Jupiter, we may want to go one step further to study how the ammonia distribution is coupled with the water convection dynamics.

4 Active 1.5-layer moist model with radiative cooling

The one-layer model of the previous section cannot be destabilized by “cooling from the top.” With a fixed saturation mixing ratio q_s and a passive layer buoyancy, radiative relaxation of the layer *thickness* only damps the height anomalies; there is no vertical buoyancy structure for the cooling to make unstable, so convection cannot spin the flow up spontaneously. We therefore promote the weather layer to a *thermally active* 1.5-layer model: a single active layer of thickness H , velocity $\mathbf{v}$, potential temperature θ and water mixing ratio q , floating on an infinitely deep, motionless abyssal layer of reference potential temperature θ_0 . The new ingredient relative to Section 2 is that the layer buoyancy is now a *prognostic* field.

Buoyancy and momentum. Define the layer buoyancy (reduced gravity)

$$b = g \frac{\theta - \theta_0}{\theta_0}. \quad (26)$$

Repeating the vertical integration of Section 1, but now retaining a horizontally varying buoyancy $b(x, y, t)$ and taking the deep layer to be at rest (reduced-gravity closure), the depth-averaged geopotential is $\langle \phi \rangle = bH$ with a baroclinic correction. Since $\phi(z) = b(H - z)$ inside the layer, the depth average of $\nabla \phi$ is $\frac{1}{2}H\nabla b + b\nabla H$, so the momentum equation becomes the thermal (Ripa) shallow-water form

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \mathbf{f} \times \mathbf{v} = -\nabla(bH) + \frac{1}{2}H \nabla b. \quad (27)$$

When $b \equiv g'$ is uniform this collapses to the usual reduced-gravity shallow water $-\nabla(g'H)$; the extra term $\frac{1}{2}H\nabla b$ is the thermal-wind forcing through which horizontal buoyancy gradients — generated by convective heating — drive the circulation.

Mass, heat and moisture. The layer thickness, heat content and water obey

$$\partial_t H + \nabla \cdot (H\mathbf{v}) = -\beta P, \quad (28)$$

$$\partial_t (Hb) + \nabla \cdot (Hb\mathbf{v}) = -H Q_R + \Lambda P, \quad (29)$$

$$\partial_t (Hq) + \nabla \cdot (Hq\mathbf{v}) = E - P. \quad (30)$$

Here Q_R is a prescribed *uniform radiative cooling* acting at the top of the layer (the direct analogue of uniform top cooling in three-dimensional convection);

E is the surface evaporation; and P is the condensation/precipitation rate. Condensation removes water mass ($-\beta P$, with $\beta = L/[C_p(\theta_2 - \theta_1)]$) as in Section 2) and releases latent heat that raises the buoyancy, with latent-buoyancy factor

$$\Lambda = \frac{Lg}{C_p \theta_0}. \quad (31)$$

Convective closure and the destabilization. Precipitation relaxes the column toward saturation once it is supersaturated, and evaporation moistens it toward a surface value,

$$P = \frac{1}{\tau_c} H(q - q_s(\theta))^+, \quad E = \frac{1}{\tau_e} H(q_{\text{surf}} - q)^+, \quad (32)$$

where $(\cdot)^+ \equiv \max(\cdot, 0)$. The essential new coupling is that the saturation mixing ratio depends on the layer temperature through Clausius–Clapeyron,

$$q_s(\theta) = q_{s0} \exp\left[\frac{L}{R_v} \left(\frac{1}{\theta_0} - \frac{1}{\theta}\right)\right] \approx q_{s0} [1 + \gamma(\theta - \theta_0)], \quad \gamma = \frac{L}{R_v \theta_0^2}. \quad (33)$$

This closes the loop that the one-layer model was missing. Uniform radiative cooling lowers θ (hence b and, through (33), q_s), which makes the column *more* prone to saturate; where the moisture supplied by evaporation is sufficient, convection fires, releasing latent heat that raises θ and q_s again and switching itself off. The resulting buoyancy anomalies drive the flow through the bH and $H\nabla b$ terms of (27). Starting from rest with $\theta = \theta_0$, $q \lesssim q_s$ and an infinitesimal seed, the cooling Q_R continually generates available potential energy that intermittent moist convection releases, spinning up turbulence and, through the β -effect, organizing it into zonal jets. The system settles into a moist radiative–convective equilibrium in which, in the global mean,

$$\underbrace{\langle Q_R \rangle}_{\text{radiative cooling}} = \frac{\Lambda}{H} \langle P \rangle, \quad \langle E \rangle = \langle P \rangle, \quad (34)$$

and the equilibrium wind amplitude is set by the imposed cooling rate Q_R rather than by the initial condition.

Prognostic variables and parameters. The active 1.5-layer model carries four prognostic fields:

- the layer thickness H and column-averaged velocity $\mathbf{v}$;
- the layer buoyancy b (equivalently the potential temperature θ);
- the column water mixing ratio q .

controlled by the parameters: the rotation $\mathbf{f}$ and gravity g ; the reference temperature θ_0 and the latent factors Λ , β ; the radiative cooling rate Q_R ; the convective and evaporative time constants τ_c , τ_e ; and the saturation curve q_{s0} , γ together with the surface value q_{surf} .